GOOGLE CLOUD × CIRCANA LIQUID DATA® | EXECUTIVE BRIEFING

Autonomous Retail Media Optimization with AlphaEvolve

Evolving Production Python Algorithms Across 5 Retail Media Networks & 30M+ Households

+165.8%

FITNESS SCORE GAIN
Seed 196.49 → Evolved 522.33

+$16.72M

INCREMENTAL SALES LIFT
+$9.90M Net-New vs. Seed ($6.82M)

6.69x

INCREMENTAL ROAS (iROAS)
Up from 2.73x on $2.50M Budget

Live Cloud Run Demo: https://circana-alpha-evolve-943928157761.us-central1.run.app

Problem Statement: Why Standard Flighting Fails

4 coupled, non-convex physical retail dynamics break standard linear & greedy budget solvers:

1. Non-Convex Hill S-Curves (Dead Zones)

- Sub-threshold spend (<0.75κ) loses page-1 sponsored auction slots and yields near-$0 lift.
- Overspending (>1.8κ) saturates the same shoppers.
- Highest marginal iROAS only occurs in the 1.15κ–1.45κ inflection knee.

2. 3-Week Household Pantry Stockpiling

- Shoppers who buy on promo stock their kitchen pantry for 2–3 weeks.
- Consecutive weekly flights on the same RMN suffer a 52% Wk+1 and 30% Wk+2 demand dip.
- Naive uniform smearing loses $7.24M to pantry fatigue.

3. Store Shelf Out-of-Stock (OOS) Ceilings

- Physical retail shelves have finite weekly supply caps (inventory_cap_musd).
- Running heavy digital promos during supply-constrained weeks truncates sales lift and incurs a 2.4x wasted-click penalty.

4. Cross-RMN Shopper Loyalty Overlap

- Circana Liquid Data® tracks 30M+ households across Walmart, Target, Kroger, Instacart & Amazon Fresh.
- 18%–35% of households shop multiple RMNs in the same week—simultaneous bursts double-pay for one cart.

Result: The naive uniform seed baseline wastes 55.7% of ad efficacy and achieves only 2.73x iROAS (Score: 196.49).

The Solution: Evolutionary Code Discovery via AlphaEvolve

AlphaEvolve treats the allocation strategy as executable Python code (# EVOLVE-BLOCK) and evolves better algorithms:

① Seed Program (initial_program.py)

- Starts with a simple 10-line proportional budget split inside # EVOLVE-BLOCK.
- Requires zero hand-crafted solver rules.
- Baseline Score: 196.49 ($6.82M Lift, 2.73x iROAS).

② Gemini Ensemble Loop (90/10 Model Mixture)

- 90% gemini-3.5-flash for rapid structural exploration.
- 10% gemini-3.1-pro / gemini-4-argon for non-convex math & DP synthesis.
- 10–20 concurrent workers mutate parent programs.

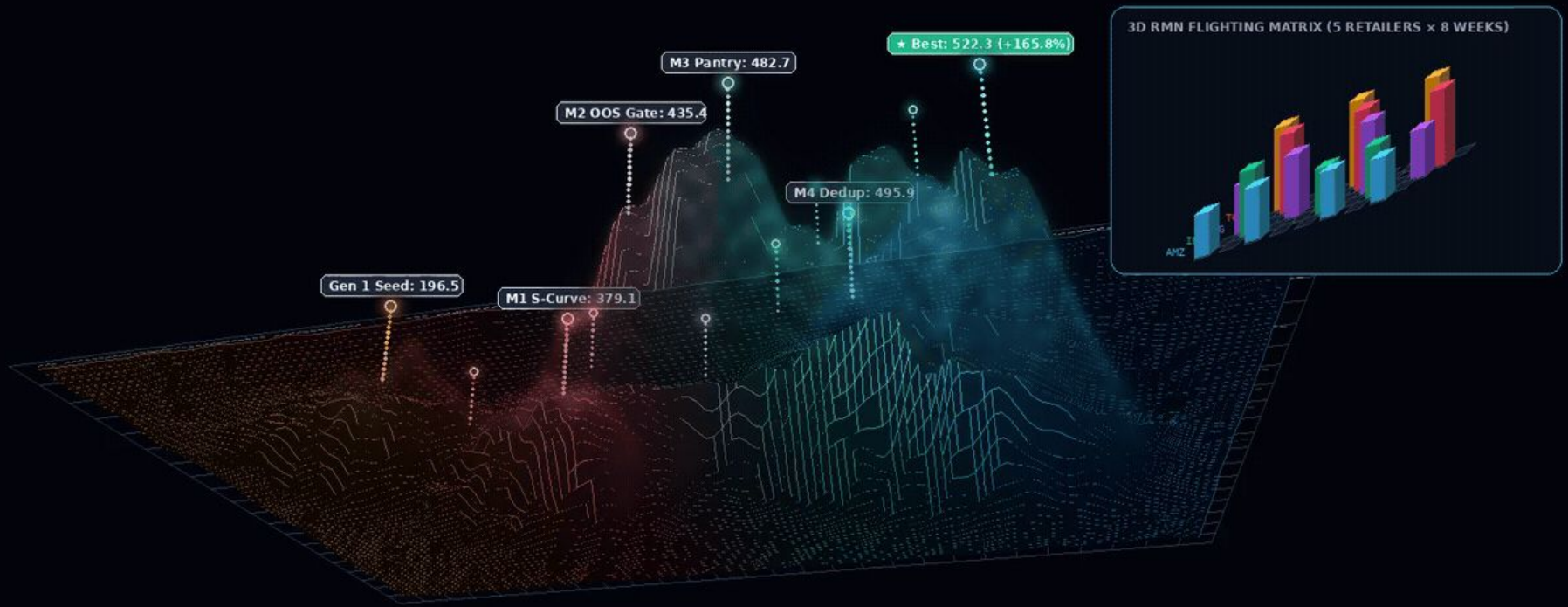
③ 3-Tier Evaluator (evaluator.py)

- Tier 1: AST security & syntax verification.
- Tier 2: 5×8 matrix shape, $2.50M cap & min-flight checks.
- Tier 3: Circana Liquid Data® simulator scoring & diagnostic feedback.

KEY ALGORITHMIC BREAKTHROUGH DISCOVERED BY ALPHAEVOLVE (nova-phoenix-100 — Score: 522.33):
Phase 1 constructs a non-colliding combinatorial pulse skeleton (staggering Walmart, Target & Kroger flights with >=2-week pantry cooldowns and OOS gating), followed by Phase 2: 140 steps of exact Marginal Hill-Elasticity Coordinate Ascent ($8k micro-transfers) to equalize marginal iROAS across all active flights.

Google Cloud AlphaEvolve Live × Circana Liquid Data®

Live 3D Evolution



Evolutionary 3D Search Space: Retail Media Flighting

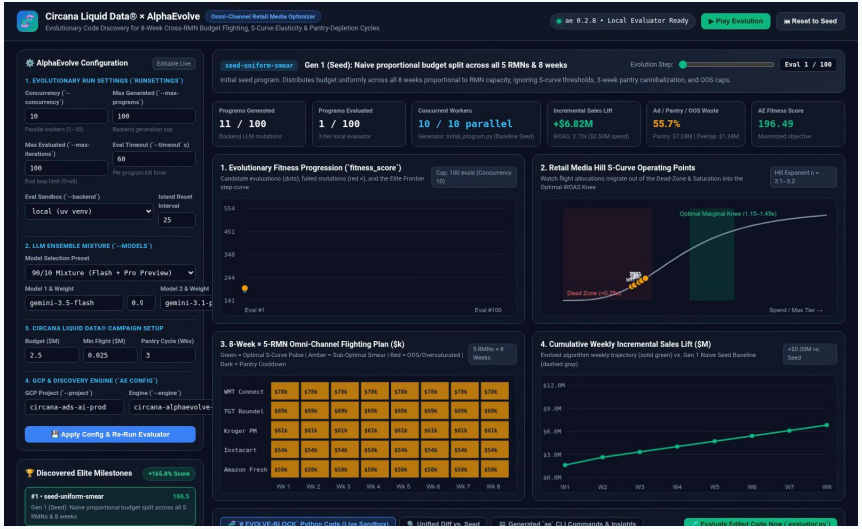
Stage [6/6] (nova-phoenix-100): Best Evolved Program: Staggered Pulse Search + Marginal Hill-Elasticity...

EVALUATOR SCORE	INCREMENTAL LIFT	CAMPAIGN iROAS	ACTIVE MODEL
522.33	$16.72M	6.69x	gemini-4-argon
+165.8% vs Seed	Net-New CPG Sales	$2.50M Ad Budget	Stage 6 / 6

```
# EVOLVE-BLOCK (nova-phoenix-100 — gemini-4-argon)
def allocate_rmn_flighting(model: Mapping[str, Any]) -> np.ndarr...
    """Best Evolved Solution: Combinatorial Pulse Selection + Ma...
    n_rmns = int(model["n_rmns"])
    n_weeks = int(model["n_weeks"])
    total_budget = float(model["total_budget_musd"])
    min_flight = float(model["min_flight_musd"])
```

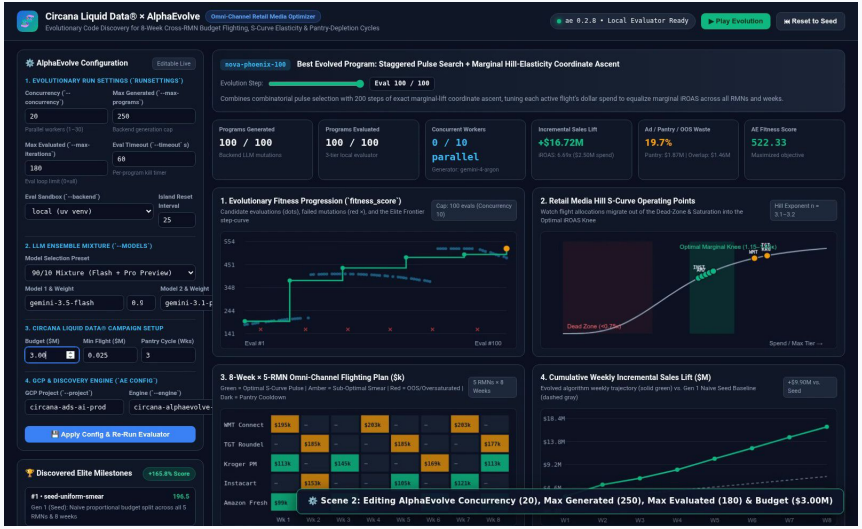
Cloud Run Studio: Naive Seed vs. Evolved Flighting Plan

BEFORE: Gen 1 Seed (Score: 196.49 | Lift: +$6.82M | 2.73x iROAS)



- Smears $50k–$78k across all 40 cells.
- Operating points stuck in the red S-curve Dead Zone (<0.75κ).
- Suffers 55.7% waste ($7.24M pantry + $1.24M overlap loss).

AFTER: Best Evolved (Score: 522.33 | Lift: +$16.72M | 6.69x iROAS)



- Concentrates $113k–$203k into staggered green S-curve pulses.
- Every flight operates inside the Optimal Marginal Knee (1.15 κ–1.45κ).
- Cuts waste to 19.7% and adds +$9.90M in net-new sales.

Cloud Run Studio: Live Code Sandbox & AE Controls

Interactive Executive & Engineering Studio

- Live AlphaEvolve Configs:
  Edit Concurrency (1–30), Max Generated, Max Evaluated, LLM Mixture Weights & Campaign Budget ($M).

- Live # EVOLVE-BLOCK Sandbox:
  Modify Python allocation code in-browser and execute evaluator.py immediately.

- Unified Diff & ae CLI Generator:
  Inspect exact code diffs vs. seed and copy ready-to-run ae CLI commands.



Public Cloud Run App: https://circana-alpha-evolve-943928157761.us-central1.run.app

Evolutionary Trajectory: 6 Discovered Milestones

Milestone / Nickname	Model	Key Algorithmic Innovation	Lift / iROAS	AE Score
Gen 1: seed-uniform-smear	Seed Baseline	Uniform weekly budget split across all 5 RMNs	$6.82M (2.73x)	**196.49**
M1: amber-falcon-18	gemini-3.5-flash	Hill S-curve inflection pulsing (1.25x kappa)	$12.44M (4.97x)	**379.11**
M2: cobalt-otter-38	gemini-3.5-flash	Store shelf Out-of-Stock (OOS) ceiling gates	$14.14M (5.66x)	**435.40**
M3: emerald-lynx-62	gemini-3.1-pro	3-week household pantry-depletion spacing	$15.55M (6.22x)	**482.74**
M4: sapphire-titan-84	gemini-4-argon	Cross-RMN household overlap deduplication	$15.94M (6.38x)	**495.88**
M5: nova-phoenix-100 ★	**gemini-4-argon**	**Pulse skeleton + 140-step marginal coordinate ascent**	**$16.72M (6.69x)**	**522.33**

+$9.90M Net-New Sales Lift

Unlocked Autonomous Code Evolution on the Exact Same $2.50M Retail Media Budget

🌐 **Live Cloud Run App**
https://circana-alpha-evolve-943928157761.us-central1.run.app
Interactive 4-chart telemetry, 3D landscape video & live Python evaluator sandbox

💻 **GitHub & AlphaEvolve CLI**
https://github.com/elhadik/Ads_Alpha_Evolve
Complete initial_program.py, 3-tier evaluator.py, offline 3D renderer & pytest suite

Why AlphaEvolve Wins for Retail & CPG Optimization:
Zero ground-up rewrite required — drop # EVOLVE-BLOCK around your existing Python heuristic, connect your simulator in evaluator.py, and let Gemini evolve verifiable, white-box production code.